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Si APD

S12023 series, etc.



Low bias operation, for 800 nm band

These are 800 nm band near-infrared Si APDs that can operate at low voltages, 200 V or less. They are suitable for applications such as FSO (free space optics) and optical rangefinders. TE-cooled type S4315 series are also avaiable (see P.6).

Features

- **Stable operation at low bias**
- **High-speed response**
- **High sensitivity and low noise**

Applications

- **FSO**
- **Optical rangefinders**

Structure / Absolute maximum ratings

Type no.	Dimensional outline/ Window material*1	Package	Effective photosensitive area size*2 (mm)	Absolute maximum ratings			
				Operating temperature*3 Topr (°C)	Storage temperature*3 Tstg (°C)	Forward current IF max (mA)	Reverse current IR max (μA)
S12023-02	(1)/K	TO-18	ϕ0.2	-20 to +85	-55 to +125	10	200
S12023-05	(1)/K		ϕ0.5				
S12051	(2)/L						
S12086	(3)/L						
S12023-10	(1)/K		ϕ1.0				
S12023-10A*4	(1)/K						
S3884	(4)/K	TO-5	ϕ1.5				
S2384	(5)/K		ϕ3.0				
S2385	(6)/K	TO-8	ϕ5.0				

*1: K=borosilicate glass, L=lens type borosilicate glass
*2: Photosensitive area in which a typical gain can be obtained
*3: No dew condensation. When there is a temperature difference between a product and the surrounding area in high humidity environments, dew condensation may occur on the product surface. Dew condensation on the product may cause deterioration in characteristics and reliability.
*4: This is a variant of the S12023-10 in which the device chip is light-shielded by aluminum layer except for the photosensitive area.
Note: Exceeding the absolute maximum ratings even momentarily may cause a drop in product quality. Always be sure to use the product within the absolute maximum ratings.

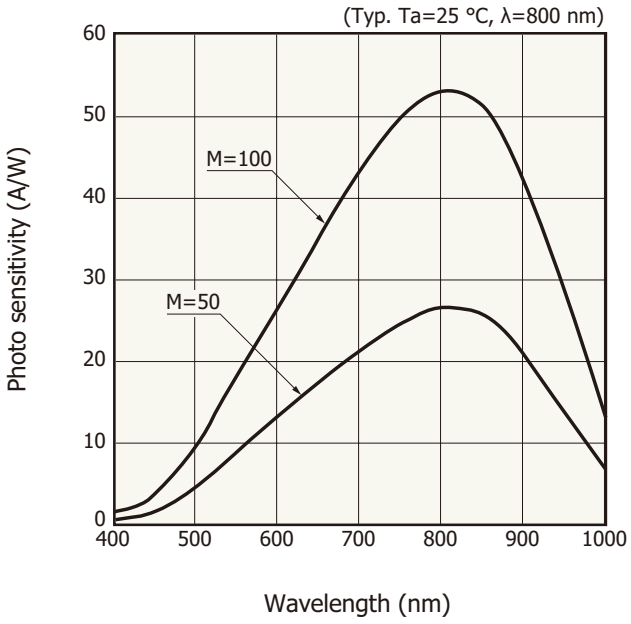
Electrical and optical characteristics (Typ. Ta=25 °C, unless otherwise noted)

Type no.	Spectral response range λ (nm)	Peak*5 sensitivity wavelength λp (nm)	Photo-sensitivity S M=1 λ=800 nm (A/W)	Quantum efficiency QE M=1 λ=800 nm (%)	Breakdown voltage VBR ID=100 µA Typ. (V)	Breakdown voltage VBR ID=100 µA Max. (V)	Temp. co-efficient of VBR (V/°C)	Dark*5 current ID Typ. (nA)	Dark*5 current ID Max. (nA)	Cutoff*5 frequency fc RL=50 Ω (MHz)	Terminal*5 capacitance Ct (pF)	Excess*5 noise figure x λ=800 nm	Gain M λ=800 nm
S12023-02	400 to 1000	800	0.5	75	150	200	0.65	0.05	0.5	1000	1	0.3	100
S12023-05								0.1	1	900	2		
S12051													
S12086													
S12023-10								0.2	2	600	6		
S12023-10A*3													
S3884								0.5	5	400	10		
S2384								1	10	120	40		60
S2385								3	30	40	95		40

*5: The value at the gain listed in "Gain M"

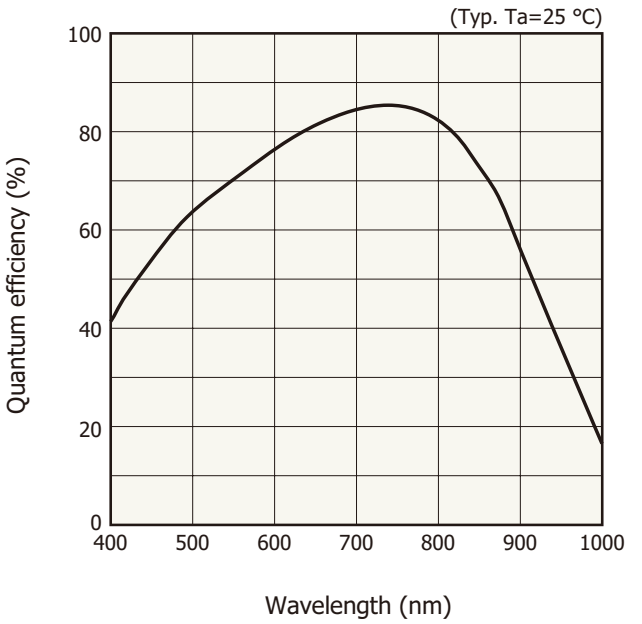
Note: Breakdown voltage can be specified by using the suffix of type number as examples shown below.
S12023-02-01: 80 to 120 V
S12023-02-02: 120 to 160 V
S12023-02-03: 160 to 200 V

Spectral response



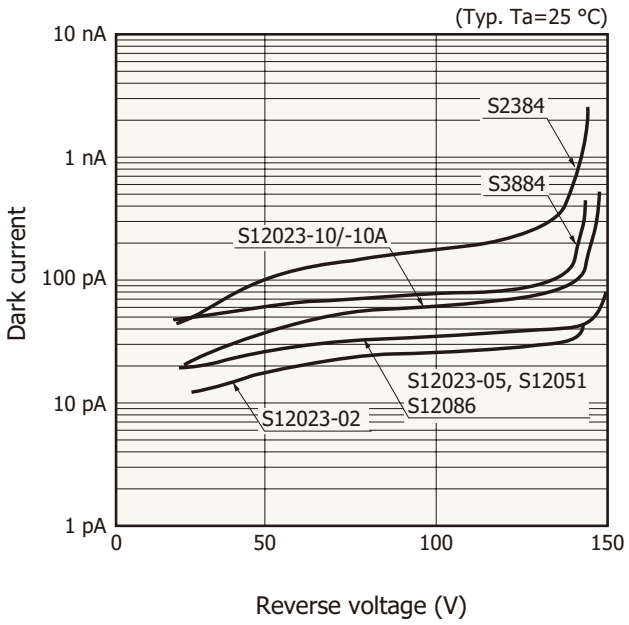
KAPDB0020ED

Quantum efficiency vs. wavelength (typical example)



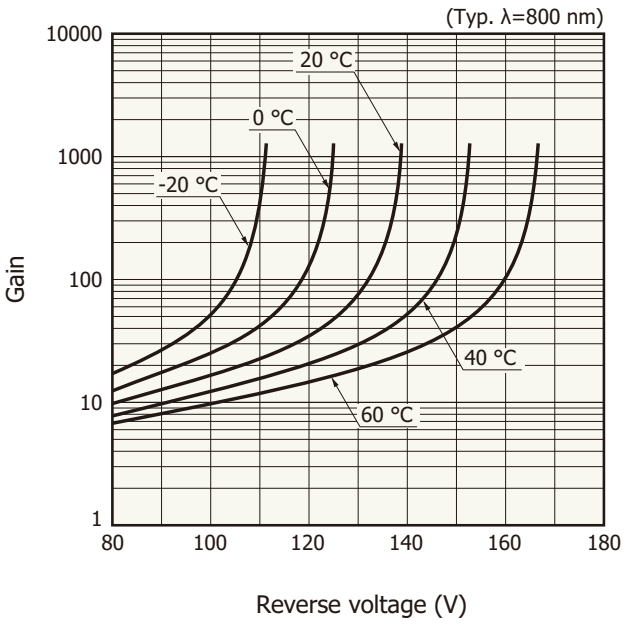
KAPDB0021EC

Dark current vs. reverse voltage



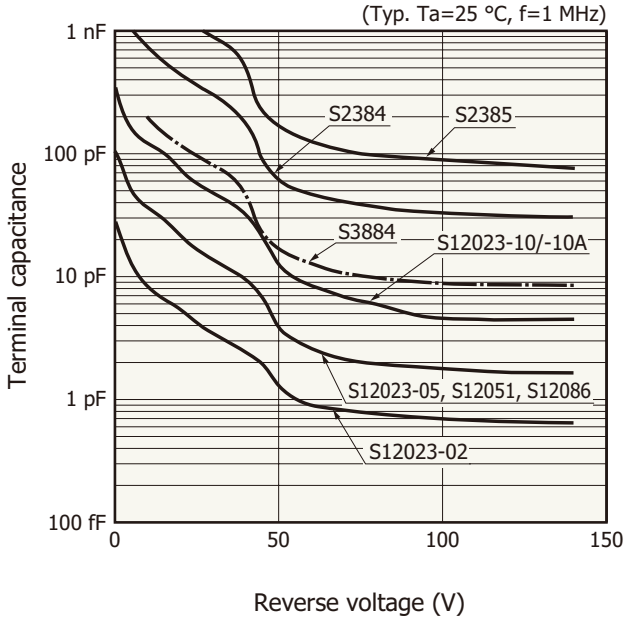
KAPDB0016ED

Gain vs. reverse voltage



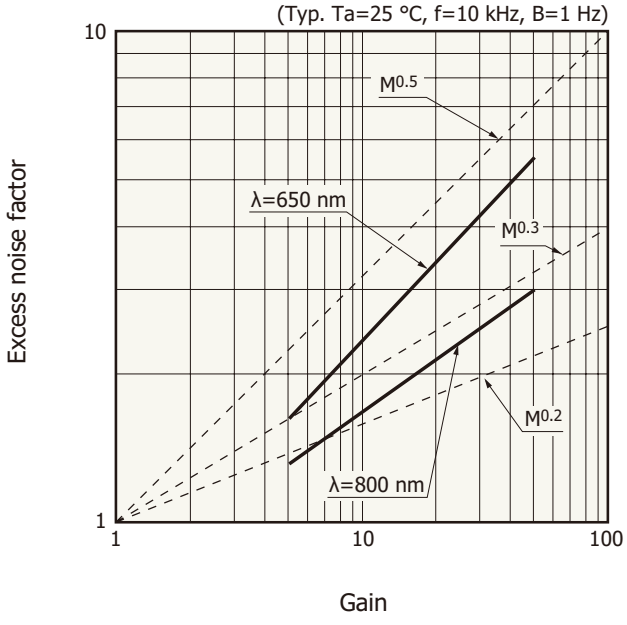
KAPDB0017EC

Terminal capacitance vs. reverse voltage



KAPDB0018ED

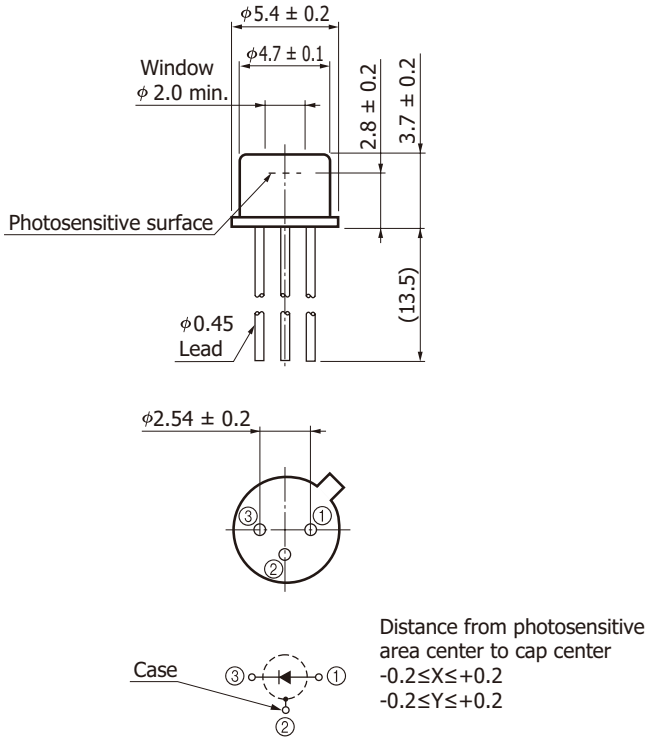
Excess noise factor vs. gain



KAPDB0022EA

Dimensional outlines (unit: mm)

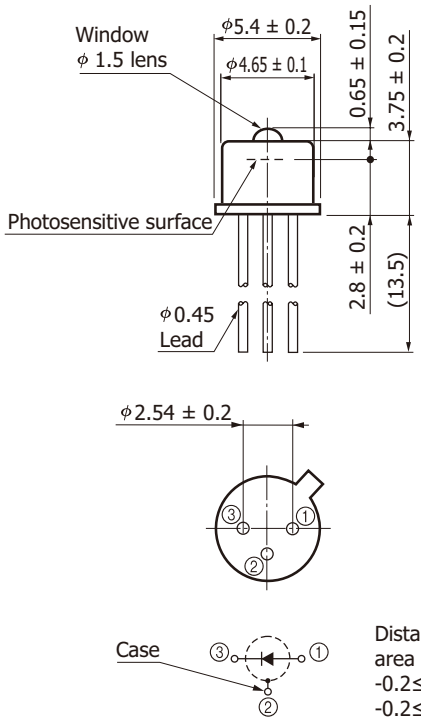
(1) S12023-02/-05/-10/-10A



Distance from photosensitive area center to cap center
-0.2≤X≤+0.2
-0.2≤Y≤+0.2

KAPDA0136EC

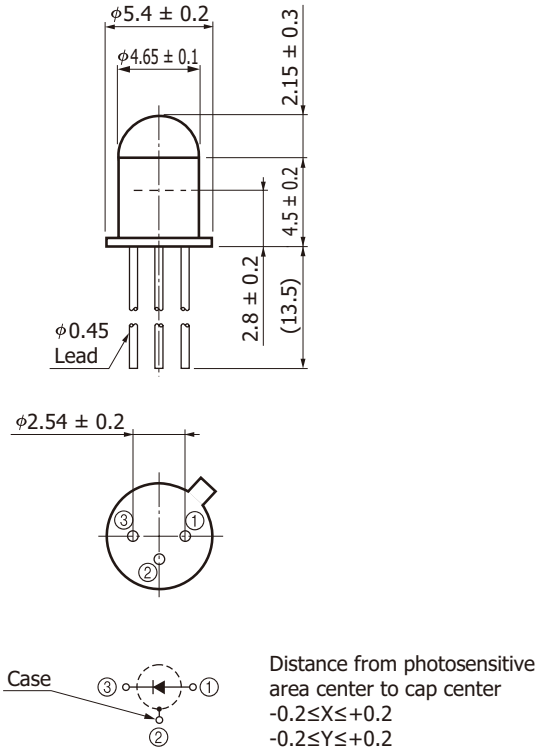
(2) S12051



Distance from photosensitive area center to cap center
-0.2≤X≤+0.2
-0.2≤Y≤+0.2

KAPDA0018EC

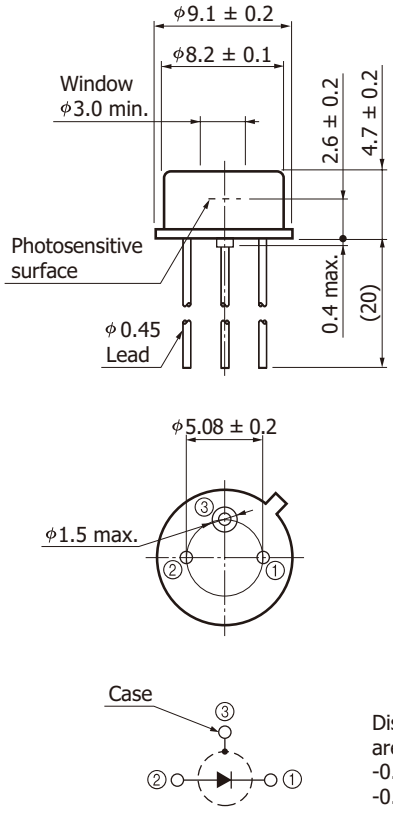
(3) S12086



Distance from photosensitive area center to cap center
-0.2≤X≤+0.2
-0.2≤Y≤+0.2

KAPDA0031EC

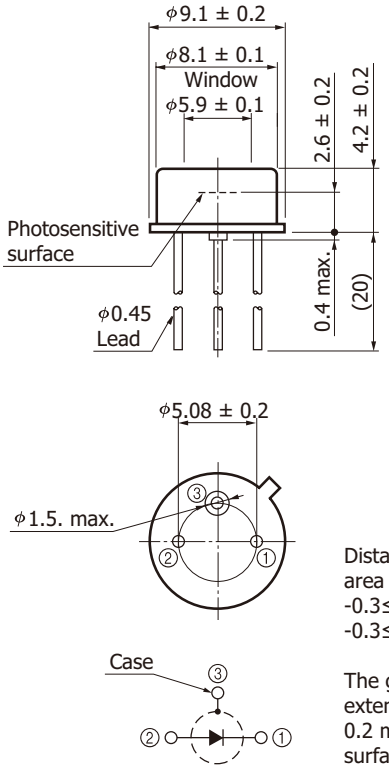
(4) S3884



Distance from photosensitive area center to cap center
-0.3≤X≤+0.3
-0.3≤Y≤+0.3

KAPDA0011EC

(5) S2384

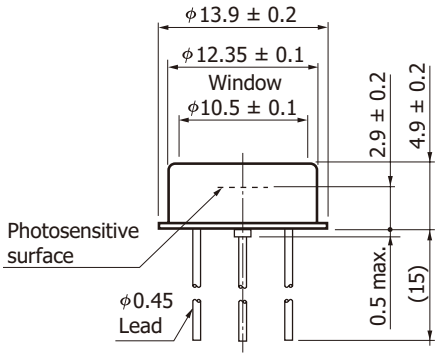


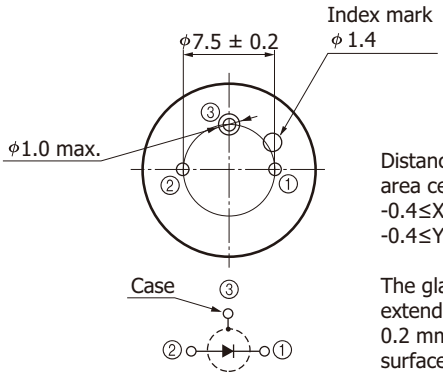
Distance from photosensitive area center to cap center
-0.3≤X≤+0.3
-0.3≤Y≤+0.3

The glass window may extend a maximum of 0.2 mm above the upper surface of the cap.

KAPDA0012EC

(6) S2385





Distance from photosensitive area center to cap center
-0.4≤X≤+0.4
-0.4≤Y≤+0.4

The glass window may extend a maximum of 0.2 mm above the upper surface of the cap.

KAPDA0013EE

Recommended soldering conditions

Solder temperature: 260 °C (10 s or less, once)
Solder the leads at a point at least 1 mm away from the package body.
Note: When you set soldering conditions, check that problems do not occur in the product by testing out the conditions in advance.

Related products

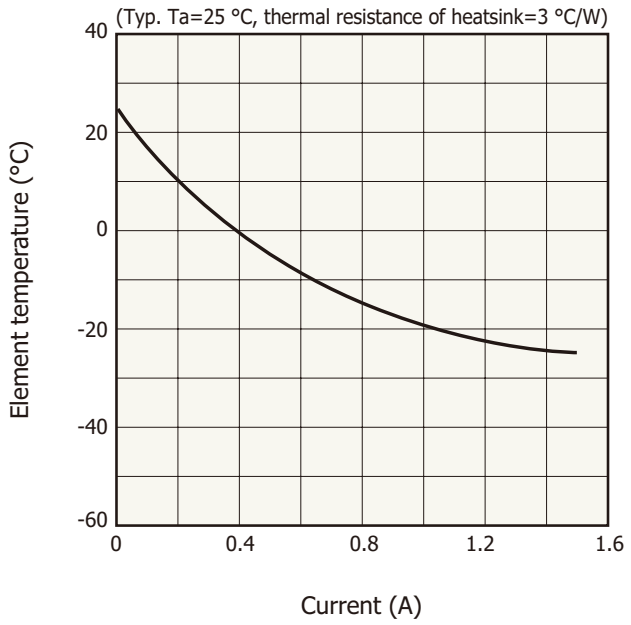
TE-cooled Si APD S4315 series

(Typ. Ta=25 °C, unless otherwise noted)

Type no.	APD used	Spectral response range λ (nm)	Peak sensitivity wavelength λp (nm)	Breakdown voltage V_{BR} I_D=100 µA Min. (V)	Typ. (V)	Max. (V)	TE-cooler allowable current I_{TE} max. (A)	TE-cooler allowable voltage V_{TE} max. (V)	Thermistor resistance Rth typ. (kΩ)	Thermistor power dissipation Pd_th max. (mW)	Operating temperature*6 Topr (°C)	Storage temperature*6 Tstg (°C)
S4315	S12023-02	400 to 1000	800*7	160	180	200	1.5	1.0	9.0	0.2	-20 to +85*9	-40 to +85
S4315-01	S12023-05											
S4315-02	S12023-10											
S4315-04	S2384		800*8									

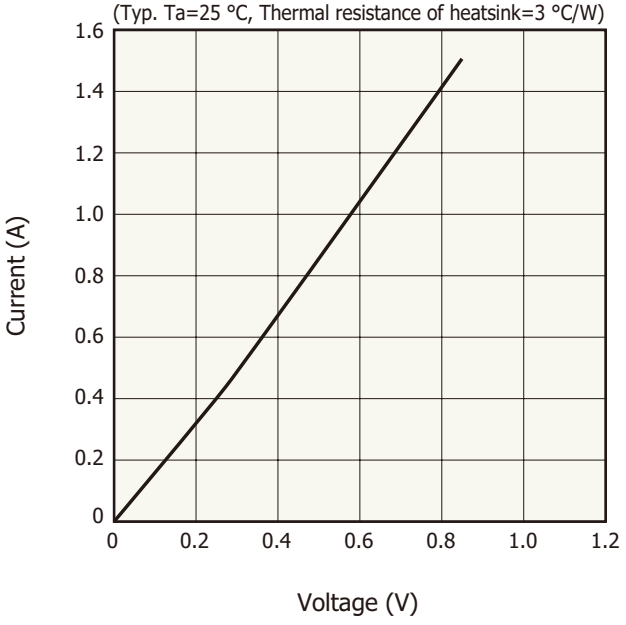
*6: No dew condensation. When there is a temperature difference between a product and the surrounding area in high humidity environments, dew condensation may occur on the product surface. Dew condensation on the product may cause deterioration in characteristics and reliability.
*7: M=100
*8: M=60
*9: Chip temperatuer and package temperature

Cooling characteristic of TE-cooler



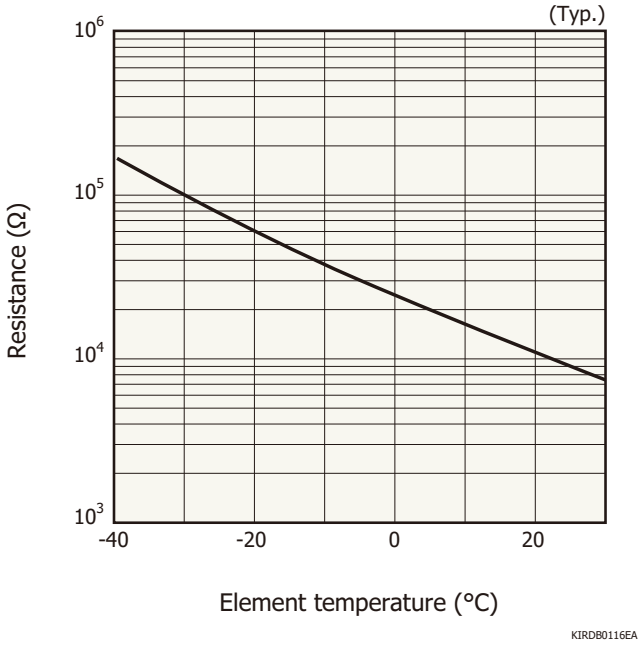
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Current vs. voltage characteristic of TE-cooler

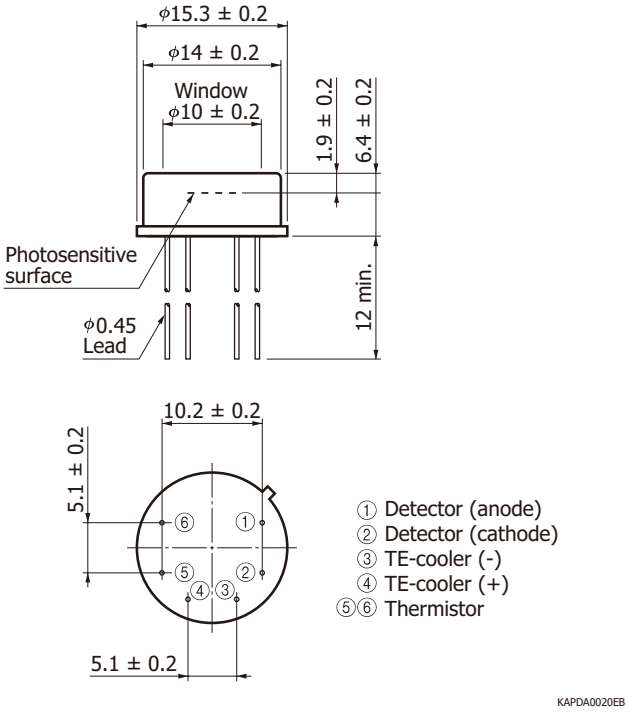


KAPDB0100EA

Thermistor temperature characteristic



Dimensional outline (unit: mm)



Recommended soldering conditions

Solder temperature: 260 °C (10 s or less, once)
Solder the leads at a point at least 1 mm away from the package body.

Related information

www.hamamatsu.com/sp/ssd/doc_en.html

■ Precautions
· Disclaimer
· Precautions / Metal, ceramic, plastic package products

■ Catalogs
· Technical note / Si APD

Information described in this material is current as of July 2026.
Product specifications are subject to change without prior notice due to improvements or other reasons. This document has been carefully prepared and the information contained is believed to be accurate. In rare cases, however, there may be inaccuracies such as text errors. Before using these products, always contact us for the delivery specification sheet to check the latest specifications.
The product warranty is valid for one year after delivery and is limited to product repair or replacement for defects discovered and reported to us within that one year period. However, even if within the warranty period we accept absolutely no liability for any loss caused by natural disasters or improper product use.
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Cat. No. KAPD1007E19
Jul. 2026 DN
Created in Japan